

Scenario Variation Module (SCVM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

Parent Standard: Simulation Scenario Standard (SSS)

Operational Layer: Simulation Scenario Governance Layer

Category: AI & Simulation

Subcategory: Simulation Governance

Type: Parent Standard Module

Governed Space: Simulation Scenario Variation

Version: 1.0

Status: Canonical · Open Module

Effective Date: 18 August 2026

Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS® · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · RIS™

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodological Intellectual Property

Canonical Language: English (UCL™)

Governed Space

Simulation Scenario Variation

Canonical Definition

Scenario Variation Module (SCVM) defines the governance framework for deliberately varying materially relevant conditions within and across identified Simulation Scenarios so that simulation behavior can be examined beyond a single nominal, expected, convenient, or fixed configuration.

SCVM governs how variation dimensions, ranges, levels, combinations, dependencies, sequences, distributions, and controlled differences are established, bounded, documented, executed as scenario configurations, and preserved for comparison.

Its purpose is to ensure that a Simulation Scenario is not treated as one static point where the governed simulation inquiry requires examination across a materially relevant range of possible conditions.

Operational Role

SCVM serves as the second internal governance space of the Simulation Scenario Standard (SSS).

The preceding module:

SCIM — Scenario Identification Module

establishes:

Which materially relevant conditions should become identifiable Simulation Scenarios?

SCVM asks:

Across which materially relevant variations of those scenarios must the simulation operate to understand how changing conditions affect behavior and outcomes?

The progression is:

Governed Scenario Register

↓

Variation Requirement Identification

↓

Variation Dimension

↓

Variation Range

↓

Variation Level / Distribution

↓

Controlled Scenario Variation

↓

Multi-Dimensional Variation

↓

Variation Dependency & Interaction

↓

Variation Configuration Set



Edge-Case Governance

SCVM therefore converts identified scenarios into a structured variation space.

Module Operational Space

SCVM governs: variation requirements, variation dimensions, variation variables, variation ranges, variation levels, continuous variation, discrete variation, categorical variation, stochastic variation, temporal variation, environmental variation, input variation, parameter variation, behavioral variation, configuration variation, dependency variation, interaction variation, sequential variation, single-factor variation, multi-factor variation, correlated variation, conditional variation, distribution-based variation, controlled comparison, reference configurations, variation boundaries, variation exclusions, variation sufficiency, variation gaps, variation traceability, variation versioning, and Variation Configuration Sets.

Variation Requirement

A Variation Requirement identifies a materially relevant dimension along which a Simulation Scenario should change.

It answers:

What can vary here in a way capable of materially changing simulation behavior or interpretation?

A Variation Requirement may arise from: Simulation Purpose, Reality Representation, Simulation Assumptions, Input Uncertainty, system operating ranges, environmental ranges, actor behavior, system configuration, timing, dependencies, resource availability, known sensitivities, or scenario-specific conditions.

Variation Dimension

A Variation Dimension is a governed characteristic across which scenario conditions may change.

Examples include:

Speed

Temperature

Power Availability

Communication Latency

Sensor Confidence

Actor Density

Human Response Time

Resource Availability

Environmental State

Autonomy Level

Failure Timing

or another simulation-specific dimension.

Variation Variable

A Variation Dimension may contain one or more variables.

For example:

Communication Condition

may contain: latency, bandwidth, packet loss, interruption duration, and connection stability.

SCVM keeps the broader condition distinct from the technical variables used to represent it.

Variation Range

A Variation Range defines the governed interval or set across which a variable may change.

Conceptually:

Minimum Relevant Condition → Expected Range → Maximum Relevant Condition

The range should derive from the Simulation Purpose and governed representation of the condition rather than arbitrary technical convenience.

Range Boundary

Every material Variation Range should establish where the intended variation begins and ends.

Where the simulation cannot represent the complete relevant range, the limitation should remain visible.

The result may be:

Required Range: X–Z

Simulation-Capable Range: X–Y

Variation Gap: Y–Z

rather than silently treating X–Y as complete coverage.

Variation Level

A continuous range does not necessarily require simulation of every possible value.

SCVM allows governed selection of representative Variation Levels.

For example:

Low

Nominal

Elevated

High

may represent one governed sampling structure.

Another inquiry may require substantially finer resolution.

The appropriate level depends on the simulation question.

Variation Resolution

Variation Resolution describes how densely a relevant range is examined.

Too little resolution may hide material behavior changes.

Excessive resolution may consume resources without materially improving understanding.

SCVM therefore asks:

At what variation resolution can materially relevant changes in behavior be detected for this Simulation Purpose?

Continuous Variation

A condition may vary continuously.

Examples include: velocity, temperature, pressure, distance, load, latency, signal strength, or resource availability.

SCVM governs how continuous ranges are bounded and sampled.

Discrete Variation

Some conditions exist as discrete states.

For example:

Mode A / Mode B / Mode C

or:

Primary System / Backup System / Emergency System

These states should not be artificially treated as continuous numerical variation where the underlying system behavior is categorical.

Categorical Variation

A scenario may vary across categories such as: environment type, actor type, vehicle class, terrain class, mission phase, population group, operational mode, or system configuration.

Material categories should remain explicitly identifiable.

Stochastic Variation

Some simulation conditions contain randomness or probabilistic behavior.

SCVM allows governed stochastic variation where appropriate.

This may include: random arrival, behavioral variability, component variation, environmental noise, uncertain timing, or probabilistic failure.

Stochastic variation should remain connected to its: distribution basis, range, assumptions, sampling method, and applicable Scenario Identity.

Distribution-Based Variation

Where a variable is represented by a distribution, the simulation should not automatically examine only its mean or most likely value.

SCVM may require examination across materially relevant portions of that distribution.

This may include:

central conditions

tails

asymmetry

multiple modes

or:

low-probability regions

where those regions materially affect the simulation inquiry.

No Mean-Only Principle

SIMULOS® establishes:

A representative average does not establish adequate variation where materially different behavior can occur away from the average.

Input Variation

Inputs governed under SIDS may vary within scenarios.

SCVM governs how those permitted inputs are configured into deliberate scenario variation.

Therefore:

SIDS → What inputs may enter

SCVM → How materially relevant input conditions should vary across scenarios
Parameter Variation

Simulation parameters may materially influence behavior.

SCVM may establish: parameter ranges, parameter levels, parameter combinations, and parameter dependencies

for scenario examination.

Parameter governance remains connected to SMES.

SCVM governs their deliberate use as scenario variation.

Environmental Variation

Simulation environments may require deliberate variation across conditions such as: weather, temperature, lighting, radiation, terrain, pressure, gravity, traffic density, network state, or another relevant environmental condition.

SCVM does not govern the environment itself.

It governs variation of environment conditions within Scenario Design.

Behavioral Variation

Where humans, AI systems, autonomous agents, organizations, or other actors participate in a simulation, their behavior may require variation.

Behavioral variation may include: reaction time, cooperation, non-cooperation, uncertainty, unexpected action, decision strategy, communication behavior, or response to system behavior.

Temporal Variation

The same event may produce materially different behavior depending upon when it occurs.

SCVM therefore recognizes:

event timing

as a potential variation dimension.

For example:

Communication Failure Before Handover

may differ materially from:

Communication Failure During Handover

or:

Communication Failure After Handover.

Sequence Variation

Where event order matters, scenario variation may require multiple sequences.

Conceptually:

$A \rightarrow B \rightarrow C$

may not be equivalent to:

$B \rightarrow A \rightarrow C$.

SCVM governs sequence variation where ordering materially changes the simulation.

Duration Variation

A condition may become material only after a certain duration.

Examples include: communication outage, resource depletion, thermal exposure, degraded sensing, delayed intervention, or sustained load.

SCVM therefore allows duration itself to become a governed variation dimension.

Rate-of-Change Variation

Not only the final value but the speed at which a condition changes may matter.

For example:

slow temperature increase

may produce different behavior from:

rapid temperature increase

even if both reach the same final value.

SCVM may therefore govern rate of change where material.

Single-Factor Variation

Single-Factor Variation changes one material condition while maintaining other relevant conditions sufficiently stable.

Conceptually:

Reference Scenario

↓

Change Variable X

↓

Observe Difference

This can help determine sensitivity to a particular variable.

Controlled Comparison

Where a simulation seeks to understand the effect of one variation, other materially relevant conditions should remain sufficiently controlled or their differences explicitly documented.

SCVM establishes:

A comparison is meaningful only to the extent that the conditions producing the difference are known.

Multi-Factor Variation

Real systems rarely experience one changing condition at a time.

SCVM therefore supports Multi-Factor Variation.

For example:

Low Power + High Communication Latency + Reduced Sensor Confidence

may represent a materially different condition from each factor independently.

Variation Interaction

Two variation dimensions may interact.

A change in Variable A may have little effect alone.

A change in Variable B may have little effect alone.

But:

$A + B$

may create materially different behavior.

SCVM therefore treats interaction as a governance question rather than assuming additive effects.

Correlated Variation

Some variables are not independent.

For example:

Severe Weather

may simultaneously affect: visibility, sensor performance, communication quality, energy consumption, and human behavior.

SCVM should preserve these relationships where independent random variation would create unrealistic scenario configurations.

No False Independence Principle

SIMULOS® establishes:

Scenario variables must not be treated as independent where material real-world dependencies make their variation structurally connected.

Conditional Variation

Some variation becomes applicable only when another condition exists.

For example:

Emergency Battery Duration

may become relevant only after:

Primary Power Loss.

SCVM allows:

Condition A → Activate Variation Dimension B

where appropriate.

Dependency Variation

Dependencies themselves may vary.

A simulation may examine: dependency available, dependency degraded, dependency intermittent, dependency delayed, dependency unavailable, or dependency recovered.

This is especially important in interconnected systems.

Configuration Variation

Different system configurations may require separate scenario variations.

Examples include: software versions, hardware configurations, sensor packages, autonomy settings, control modes, or infrastructure configurations.

Materially different configurations should remain traceable to the scenario variation in which they are used.

Variation Combination

Variation dimensions may be combined into a Variation Configuration.

Conceptually:

Scenario Identity Input Condition Environment Parameter State Behavioral
Condition Timing

=

Variation Configuration Variation

Configuration Identity

Every materially distinct Variation Configuration should be sufficiently identifiable.

It may include: Scenario ID, Variation Configuration ID, variable values, categorical states, applicable ranges, dependency conditions, sequence, timing, environment configuration, parameter configuration, input configuration, and version.

Variation Configuration Set

A Variation Configuration Set is the governed collection of variation configurations selected for a Scenario or Scenario Family.

It represents the condition space that subsequent simulation activities may execute.

Combinatorial Explosion

Complex simulations may contain hundreds or thousands of possible variation dimensions.

Their combinations may become effectively unlimited.

SCVM therefore does not require:

every variable × every value × every other variable

to be simulated.

Instead it requires governed selection of combinations capable of materially improving examination of the Simulation Purpose.

Variation Prioritization

Where exhaustive variation is impractical, prioritization may consider: materiality, uncertainty, sensitivity, consequence, dependency, interaction potential, boundary proximity, historical behavior, operational relevance, novelty, and downstream reliance.

Variation Sufficiency

Variation Sufficiency asks:

Has the scenario been varied sufficiently to expose materially different behavior that could reasonably occur within the governed simulation inquiry?

This is not a universal numerical threshold.

It is a governed sufficiency determination relative to: Simulation Purpose, Scenario Requirement, represented reality, uncertainty, consequence, and available simulation capability.

Variation Gap

A Variation Gap exists where a materially relevant range, category, dependency, interaction, sequence, or configuration has not been sufficiently represented.

A gap may result from: unavailable data, model limitations, computational constraints, unknown distributions, insufficient scenario resolution, excluded conditions, or undiscovered dependencies.

No Capability-Erasure Principle

SIMULOS® establishes:

A simulation limitation does not eliminate the variation requirement that exposed the limitation.

Where required variation cannot be simulated, the gap should remain explicit.

Variation Exclusion

A potential variation may be excluded where it is: outside Simulation Scope, materially impossible, irrelevant to the Scenario Requirement, redundant, or sufficiently represented by another governed configuration.

Material exclusions should be documented where omission could affect interpretation.

No Convenience-Range Principle

A Variation Range should not be narrowed solely because: available data cover only part of it, the model performs poorly outside a comfortable region, computation becomes expensive, results become unstable, or unfavorable behavior appears.

These may justify a limitation.

They do not justify hiding the limitation.

Variation Traceability

SCVM should preserve relationships such as:

Scenario Requirement

↓

Scenario ID

↓

Variation Requirement

↓

Variation Dimension

↓

Variation Range

↓

Variation Configuration

↓

Simulation Execution

This allows a simulation result to be traced back to the exact variation conditions under which it occurred.

Variation Versioning

Variation definitions may change.

Material changes to: ranges, distributions, categories, parameter values, dependencies, or configuration logic

should remain version-distinguishable.

Variation Change Trigger

Reassessment may be required where: new operational evidence appears, system configuration changes, environmental understanding changes, new dependencies are discovered, assumptions change, input distributions change, or previous simulations expose new sensitivities.

Adaptive Variation Expansion

Simulation execution may reveal that a particular region of the variation space produces unusual behavior.

SCVM may support:

Initial Variation → Observed Sensitivity → Increased Variation Resolution → Additional Configurations

This allows simulation resources to be concentrated where new information is being discovered.

No Static Sampling Principle

SIMULOS® establishes:

A predefined variation plan should not prevent expansion when simulation behavior reveals a materially important region that was not sufficiently examined.

Variation and Uncertainty

Variation and uncertainty are related but not identical.

Uncertainty asks:

What do we not know sufficiently?

Variation asks:

Across which different conditions should the simulation deliberately operate?

Uncertainty may create a Variation Requirement.

But not every variation represents uncertainty.

Some variation represents known operational diversity.

Variation and Edge Cases

SCVM governs the structured variation of scenarios.

The following module:

SECM — Scenario Edge-Case Module

examines conditions near unusual, sparse, boundary, extreme, or otherwise materially atypical regions of that variation space.

Therefore:

SCVM maps the variation space.

SECM interrogates its difficult edges.

Variation and Stress Conditions

SCVM establishes ordinary and materially relevant variation.

SSCM — Scenario Stress-Condition Module subsequently governs deliberate stress beyond ordinary scenario conditions where required.

Variation should therefore not be inflated to absorb the separate governance purpose of stress testing.

Variation and Coverage

Variation configurations become inputs to:

SCCM — Scenario Coverage Module.

Coverage asks whether the selected variations collectively examine the materially relevant Scenario Space.

SCVM does not make the final Scenario Coverage determination.

Space and Mission Systems

Space systems demonstrate why controlled variation matters.

A mission simulation may examine a spacecraft under: different communication delays, changing power availability, navigation uncertainty, thermal conditions, radiation environments, timing deviations, subsystem degradation, and mission-phase configurations.

The important question is not simply whether each variable was simulated.

It may be whether materially relevant combinations and timing relationships were examined.

For example:

Reduced Power + Communication Delay + Navigation Uncertainty during Autonomous Recovery

may expose behavior not visible in isolated variation.

SCVM provides the structure for governing those combinations without prescribing which combinations every mission must simulate.

Autonomous Systems

Autonomous-system behavior can depend strongly on simultaneous variation in: perception quality, environmental complexity, actor behavior, communication, resource availability, autonomy state, decision timing, and system confidence.

A system that performs well when these dimensions are varied individually may behave differently when several change together.

SCVM therefore provides a structured way to move from:

“We tested many conditions.”

to:

“We can show which dimensions changed, across what ranges, in which combinations, and why those variations were selected.” AI and Multi-Agent Systems

Multi-agent environments can create especially large variation spaces because each agent may introduce: behavior variation, objective variation, timing variation, communication variation, strategy variation, and interaction variation.

SCVM does not require exhaustive enumeration.

It requires that material variation and excluded variation remain visible and governable.

Industrial Simulation

An industrial system may perform correctly at:

normal load + normal temperature + full sensor availability

while becoming unstable under:

high load + elevated temperature + partial sensor degradation.

SCVM makes the difference between these configurations explicit and traceable.

Existing Simulation Program Diagnostic

SCVM can be applied to an existing simulation program.

An organization can ask:

Which variables do our scenarios actually vary?

Across what ranges?

Why were those ranges selected?

Which dimensions are held constant?

Which variables are correlated?

Which combinations have never been examined?

Are we testing distributions or only averages?

Where are our Variation Gaps?

This can reveal that a simulation program has high execution volume but low variation diversity.

Module Boundary

SCVM begins when SCIM has established sufficiently identifiable Simulation Scenarios and Scenario Requirements.

SCVM ends when materially relevant variation dimensions, ranges, levels, dependencies, interactions, combinations, and configurations have been sufficiently governed to form a Variation Configuration Set suitable for Edge-Case, Stress-Condition, and Coverage governance.

SCVM does not govern: general Simulation Input governance, general model parameter governance, complete edge-case identification, stress-condition design, final Scenario Coverage, Simulation Execution, or simulation-output interpretation.

It governs specifically:

how identified scenarios are deliberately varied across materially relevant conditions.

Minimum Implementation Framework

1. Identify Variation Requirements

Determine which conditions within each Scenario can materially vary. 2. Define Variation Dimensions

Identify: variables, categories, states, timing, sequences, environmental conditions, behavioral conditions, and dependencies.

3. Establish Variation Ranges

Define materially relevant: minimums, maximums, categories, distributions, and applicability boundaries.

4. Establish Variation Structure

Determine whether variation is: continuous, discrete, categorical, stochastic, temporal, sequential, single-factor, multi-factor, correlated, or conditional.

5. Construct Variation Configurations

Create governed combinations suitable for simulation. 6. Assess Variation

Sufficiency

Determine whether materially relevant ranges and relationships are sufficiently represented. 7. Record Variation Gaps and Exclusions

Preserve conditions that remain unexamined or cannot currently be represented.
8. Establish Variation Configuration Set

Produce the governed variation basis for downstream Scenario Design.

Governance Outputs

SCVM may produce: Variation Requirement, Variation Dimension Register, Variation Variable Record, Variation Range, Variation Boundary, Variation Level, Variation Resolution Record, Continuous Variation Record, Discrete Variation Record, Categorical Variation Record, Stochastic Variation Record, Distribution Basis, Input Variation Record, Parameter Variation Record, Environmental Variation Record, Behavioral Variation Record, Temporal Variation Record, Sequence Variation Record, Duration Variation Record, Rate-of-Change Variation Record, Single-Factor Variation Record, Multi-Factor Variation Record, Correlated Variation Record, Conditional Variation Record, Dependency Variation Record, Configuration Variation Record, Variation Interaction Map, Variation Configuration, Variation Configuration Identifier, Variation Configuration Set, Variation Prioritization Record, Variation Sufficiency Assessment, Variation Gap, Variation Exclusion Record, Variation Traceability Map, Variation Version Record, and Governed Scenario Variation Basis.

Use Case 1 — Autonomous Driving

Scenario

An autonomous-driving simulation contains a pedestrian-crossing scenario.

Testing initially varies only vehicle speed.

SCVM identifies additional materially relevant Variation Dimensions: visibility, pedestrian speed, pedestrian direction, occlusion, road surface, sensor confidence, reaction timing, surrounding traffic, and pedestrian behavior.

Application

The program first examines controlled single-factor variation.

It then identifies materially relevant interactions, including:

Low Visibility + Partial Occlusion + Reduced Sensor Confidence + Late Pedestrian Entry

rather than treating each variable as independent. Result

One static pedestrian scenario becomes a governed variation space.

The program can identify which combinations were examined and which remain Variation Gaps. Use Case 2 — Spacecraft Autonomous Recovery

Scenario

A spacecraft simulation includes autonomous recovery after loss of nominal navigation.

The existing program tests the recovery under one standard communication and power configuration.

SCVM identifies material variation in: navigation uncertainty, communication delay, communication interruption duration, available power, thermal condition, recovery initiation timing, and subsystem availability.

Application

The variables are not treated as universally independent.

Reduced power affects which subsystems remain available.

Communication condition affects ground-intervention opportunity.

Recovery timing interacts with mission phase.

SCVM establishes governed Variation Configurations reflecting those relationships. Result

The organization can distinguish:

Recovery Was Simulated

from the stronger and more useful statement:

Recovery was examined across these defined operating ranges, dependencies, timing conditions, and materially relevant combinations. Use Case 3 — Industrial Autonomous Robot

Scenario

An autonomous industrial robot operates safely during simulations with normal production load.

The facility wants to understand performance under changing operational conditions. Application

SCVM establishes variation across: production density, human proximity, sensor confidence, payload, floor conditions, communication latency, actuator performance, and human reaction timing.

Simulation results reveal unusual behavior only when:

High Production Density + Reduced Sensor Confidence + Increased Payload

occur together. Result

The variation program exposes a system behavior that would not have appeared through isolated one-variable testing.

The finding can then generate new Edge-Case or Stress-Condition requirements without altering the purpose of SCVM itself.

Architectural Position

Scenario Variation Module (SCVM) is the second internal governance space of the Simulation Scenario Standard (SSS).

The complete SSS progression is:

Scenario Identification

↓

Scenario Variation

↓

Scenario Edge Cases

↓

Scenario Stress Conditions

↓

Scenario Coverage

SCIM establishes which scenarios matter.

SCVM establishes how those scenarios must vary.

SECM determines which difficult edges and atypical conditions require deliberate examination.

SSCM governs material stress conditions.

SCCM determines whether the resulting Scenario Space is sufficiently covered.

Foundational Principle

A scenario examined at only one convenient configuration cannot demonstrate behavior across a condition space that materially varies.

Canonical Closing Statement

Scenario Variation Module transforms identified Simulation Scenarios from static test conditions into governed variation spaces by defining which dimensions may materially change, across what ranges, distributions, states, sequences, dependencies, and combinations those changes must be examined, and how resulting configurations remain identifiable and traceable. Through controlled single-factor variation, multi-factor interaction, correlated

conditions, temporal and sequence variation, variation sufficiency, explicit gaps, exclusions, adaptive expansion, and complete configuration traceability, SCVM ensures that simulation programs do not mistake testing one representative condition—or many repetitions of similar conditions—for systematic examination of the materially relevant variation space surrounding the Simulation Object.

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>